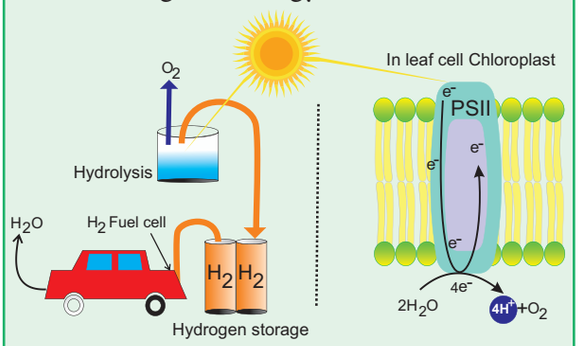
**A quest for future energy**

Hydrogen is considered as a promising energy vector for the next generation. It can be used for "green" electricity production or developing cogeneration systems such as fuel cells. The sustainability of its employment depends on the energy source used to synthesize it from hydrogen-rich compounds such as water or biomass. The splitting of water in hydrogen and oxygen by means of solar radiation in Photolysis is common in plants. Water splitting is not an easy process to mimic artificially but preliminary success is achieved so far. If young minds take up this as their research ambition a revolution can be made in green energy.




major endergonic reaction. In this chapter, we will study about the energy yielding process of photosynthesis and various types of energy utilization processes to produce carbohydrates.

## 13.1 Definition, Significance and Site of Photosynthesis

### 13.1.1 Definition of Photosynthesis

Photosynthesis is referred as photochemical oxidation and reduction reactions carried out with the help of light, converting solar energy into chemical energy. It is the most important anabolic process. Plants and photosynthetic bacteria use simple raw materials like carbon dioxide water and with the help of light energy synthesize carbohydrates and evolve oxygen. The overall chemical equation for photosynthesis is:

$$6CO_2 + 6H_2O \xrightarrow[\text{Chlorophyll}]{\text{Light}} C_6H_{12}O_6 + 6O_2\uparrow$$

**Ruben** and **Kamen** (1941) demonstrated six molecules of water as insufficient for the evolution of 6 molecules of $O_2$ and modified the equation as:

$$6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{Light}} C_6H_{12}O_6 + 6H_2O + 6O_2\uparrow$$

Photosynthesis is a collection of oxidation and reduction reactions (Redox reaction).

**Oxidation**- Water is oxidised into oxygen (loss of electrons).

**Reduction** – $CO_2$ is reduced into Carbohydrates (gain of electrons).

In some bacteria, oxygen is not evolved and is called as **non-oxygenic** and **anaerobic photosynthesis**. Examples: Green sulphur, Purple sulphur and green filamentous bacteria.

### 13.1.2 Significance of Photosynthesis

1. Photosynthetic organisms provide food for all living organisms on earth either directly or indirectly.
2. It is the only natural process that liberates oxygen in the atmosphere and balances the oxygen level.
3. Photosynthesis balances the oxygen and carbon cycle in nature.
4. Fuels such as coal, petroleum and other fossil fuels are from preserved photosynthetic plants.
5. Photosynthetic organisms are the primary producers on which all consumers depend for energy.
6. Plants provide fodder, fibre, fire wood, timber, useful medicinal products and these sources come by the act of photosynthesis.

### 13.1.3 Site of Photosynthesis

Chloroplasts are the main site of photosynthesis and both energy yielding process (Light reaction) and fixation of carbon di oxide (Dark reaction)that takes place in chloroplast. It is a double wall membrane bounded organelle, discoid or lens shaped, 4–10 µm in diameter and 1–33 µm in thickness. The membrane is a unit membrane and space between them is 100 to 200 A°. A colloidal and proteinaceous matrix called stroma is present inside.



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